

- 6.10. Specify the following queries in SQL on the COMPANY relational database schema shown in Figure 5.5. Show the result of each query if it is applied to the COMPANY database in Figure 5.6.
- Retrieve the names of all employees in department 5 who work more than 10 hours per week on the ProductX project.
  - List the names of all employees who have a dependent with the same first name as themselves.
  - Find the names of all employees who are directly supervised by 'Franklin Wong'.

**EMPLOYEE**

|       |       |       |            |       |         |     |        |           |     |
|-------|-------|-------|------------|-------|---------|-----|--------|-----------|-----|
| Fname | Minit | Lname | <u>Ssn</u> | Bdate | Address | Sex | Salary | Super_ssn | Dno |
|-------|-------|-------|------------|-------|---------|-----|--------|-----------|-----|

**DEPARTMENT**

|       |                |         |                |
|-------|----------------|---------|----------------|
| Dname | <u>Dnumber</u> | Mgr_ssn | Mgr_start_date |
|-------|----------------|---------|----------------|

**DEPT\_LOCATIONS**

|                |                  |
|----------------|------------------|
| <u>Dnumber</u> | <u>Dlocation</u> |
|----------------|------------------|

**PROJECT**

|       |                |           |      |
|-------|----------------|-----------|------|
| Pname | <u>Pnumber</u> | Plocation | Dnum |
|-------|----------------|-----------|------|

**WORKS\_ON**

|             |            |       |
|-------------|------------|-------|
| <u>Essn</u> | <u>Pno</u> | Hours |
|-------------|------------|-------|

**DEPENDENT**

|             |                       |     |       |              |
|-------------|-----------------------|-----|-------|--------------|
| <u>Essn</u> | <u>Dependent_name</u> | Sex | Bdate | Relationship |
|-------------|-----------------------|-----|-------|--------------|

**Figure 5.5**  
Schema diagram for the  
COMPANY relational  
database schema.

- 6.12. Specify the following queries in SQL on the database schema of Figure 1.2.
- Retrieve the names of all senior students majoring in 'cs' (computer science).
  - Retrieve the names of all courses taught by Professor King in 2007 and 2008.
  - For each section taught by Professor King, retrieve the course number, semester, year, and number of students who took the section.
  - Retrieve the name and transcript of each senior student (Class = 4) majoring in CS. A transcript includes course name, course number, credit hours, semester, year, and grade for each course completed by the student.

**STUDENT**

| Name  | Student_number | Class | Major |
|-------|----------------|-------|-------|
| Smith | 17             | 1     | CS    |
| Brown | 8              | 2     | CS    |

**COURSE**

| Course_name               | Course_number | Credit_hours | Department |
|---------------------------|---------------|--------------|------------|
| Intro to Computer Science | CS1310        | 4            | CS         |
| Data Structures           | CS3320        | 4            | CS         |
| Discrete Mathematics      | MATH2410      | 3            | MATH       |
| Database                  | CS3380        | 3            | CS         |

**SECTION**

| Section_identifier | Course_number | Semester | Year | Instructor |
|--------------------|---------------|----------|------|------------|
| 85                 | MATH2410      | Fall     | 07   | King       |
| 92                 | CS1310        | Fall     | 07   | Anderson   |
| 102                | CS3320        | Spring   | 08   | Knuth      |
| 112                | MATH2410      | Fall     | 08   | Chang      |
| 119                | CS1310        | Fall     | 08   | Anderson   |
| 135                | CS3380        | Fall     | 08   | Stone      |

**GRADE\_REPORT**

| Student_number | Section_identifier | Grade |
|----------------|--------------------|-------|
| 17             | 112                | B     |
| 17             | 119                | C     |
| 8              | 85                 | A     |
| 8              | 92                 | A     |
| 8              | 102                | B     |
| 8              | 135                | A     |

**PREREQUISITE**

| Course_number | Prerequisite_number |
|---------------|---------------------|
| CS3380        | CS3320              |
| CS3380        | MATH2410            |
| CS3320        | CS1310              |

**Figure 1.2**  
A database that stores student and course information.